AERIS Expert Review Packet

Anomaly

Source / metric	tceq / co
Observed / expected	0.5 / 0.136957
Time	2026-07-29 17:00 UTC
Location	29.7337, -95.2576
Severity	severe
Detectors	zscore, stl, isolation_forest

Stored evidence summary

Source	Metric	Unit	Entities / points	Range; mean	Nearest to event
asos	humidity	percent	9 / 865	40.37 to 100; mean 72.9254	59.12 at 2026-07-29 16:55Z; dt 5 min
asos	temperature	degC	9 / 865	22.7778 to 36.1111; mean 29.651	32 at 2026-07-29 16:55Z; dt 5 min
asos	wind_direction	degree	9 / 1160	0 to 270; mean 163.543	190 at 2026-07-29 16:55Z; dt 5 min
asos	wind_speed	m/s	9 / 1212	0 to 7.71666; mean 2.8498	2.05778 at 2026-07-29 16:55Z; dt 5 min
noaa_gfs	gh_500	m	10 / 120	5909.24 to 5941.31; mean 5927.82	5928.39 at 2026-07-29 18:00Z; dt 60 min
noaa_gfs	pbl_height	m	10 / 120	48.9097 to 2294.53; mean 860.215	2188.77 at 2026-07-29 18:00Z; dt 60 min
noaa_gfs	precipitable_water	mm	10 / 120	34.4259 to 49.1773; mean 40.4243	43.2396 at 2026-07-29 18:00Z; dt 60 min
noaa_gfs	surface_pressure	hPa	10 / 120	1004.76 to 1014.21; mean 1009.87	1010.7 at 2026-07-29 18:00Z; dt 60 min
noaa_gfs	t_850	degC	10 / 120	18.3487 to 22.1724; mean 20.4638	19.1542 at 2026-07-29 18:00Z; dt 60 min
noaa_gfs	u_10m	m/s	10 / 120	-3.50088 to 2.79912; mean 0.6582	1.74827 at 2026-07-29 18:00Z; dt 60 min
noaa_gfs	v_10m	m/s	10 / 120	0.334747 to 6.23475; mean 3.0269	3.18474 at 2026-07-29 18:00Z; dt 60 min
openaq	ozone	ppm	17 / 1158	-0.004 to 0.053; mean 0.0177	0.018 at 2026-07-29 17:00Z; dt 0 min
openaq	pm10	ug/m3	3 / 216	16 to 143; mean 41.4167	120 at 2026-07-29 17:00Z; dt 0 min
openaq	pm25	ug/m3	69 / 2909	0 to 37.5; mean 6.2519	15.7 at 2026-07-29 17:00Z; dt 0 min
openweather	cloud_cover	percent	5 / 360	0 to 100; mean 39.7528	33 at 2026-07-29 17:01Z; dt 1.9 min
openweather	humidity	percent	5 / 360	49 to 93; mean 74.7111	69 at 2026-07-29 17:01Z; dt 1.9 min
openweather	precipitation	mm/h	5 / 360	0 to 0; mean 0	0 at 2026-07-29 17:01Z; dt 1.9 min
openweather	pressure	hPa	5 / 360	1009 to 1016; mean 1012.32	1013 at 2026-07-29 17:01Z; dt 1.9 min
openweather	temperature	degC	5 / 360	24.98 to 36.85; mean 30.463	33.72 at 2026-07-29 17:01Z; dt 1.9 min
openweather	wind_direction	degree	5 / 360	0 to 321; mean 192.231	208 at 2026-07-29 17:01Z; dt 1.9 min
openweather	wind_speed	m/s	5 / 360	0.45 to 6.55; mean 2.2293	2.42 at 2026-07-29 17:01Z; dt 1.9 min
purpleair	pm25	ug/m3	74 / 5222	0 to 232; mean 8.0665	8.3 at 2026-07-29 17:00Z; dt 0 min

Source	Metric	Unit	Entities / points	Range; mean	Nearest to event
sentinel5p	s5p_aer_ai_granule_available	count	9 / 9	1 to 1; mean 1	1 at 2026-07-29 18:13Z; dt 73.8 min
sentinel5p	s5p_ch4_granule_available	count	9 / 9	1 to 1; mean 1	1 at 2026-07-29 18:13Z; dt 73.8 min
sentinel5p	s5p_co_column	mol/m^2	9 / 9	0.0256055 to 0.0278855; mean 0.0271	0.0256055 at 2026-07-29 18:13Z; dt 73.8 min
sentinel5p	s5p_co_granule_available	count	9 / 9	1 to 1; mean 1	1 at 2026-07-29 18:13Z; dt 73.8 min
sentinel5p	s5p_hcho_column	mol/m^2	7 / 7	0.000175431 to 0.0002266; mean 0.0002	0.000223665 at 2026-07-29 18:13Z; dt 73.8 min
sentinel5p	s5p_hcho_granule_available	count	7 / 7	1 to 1; mean 1	1 at 2026-07-29 18:13Z; dt 73.8 min
sentinel5p	s5p_no2_column	mol/m^2	5 / 5	3.65242e-05 to 4.17212e-05; mean 0	4.17212e-05 at 2026-07-29 18:44Z; dt 104.6 min
sentinel5p	s5p_no2_granule_available	count	5 / 5	1 to 1; mean 1	1 at 2026-07-29 18:44Z; dt 104.6 min
sentinel5p	s5p_o3_granule_available	count	9 / 9	1 to 1; mean 1	1 at 2026-07-29 18:13Z; dt 73.8 min
sentinel5p	s5p_so2_column	mol/m^2	7 / 7	-7.46696e-06 to 0.000148654; mean 0.0001	0.000148654 at 2026-07-29 18:13Z; dt 73.8 min
sentinel5p	s5p_so2_granule_available	count	7 / 7	1 to 1; mean 1	1 at 2026-07-29 18:13Z; dt 73.8 min
tceq	co	ppm	3 / 188	0.1 to 0.8; mean 0.2069	0.5 at 2026-07-29 17:00Z; dt 0 min
tceq	no2	ppb	12 / 707	-2.6 to 19.7; mean 4.2532	10.5 at 2026-07-29 17:00Z; dt 0 min
tceq	so2	ppb	3 / 190	-0.7 to 5.7; mean -0.0842	-0.1 at 2026-07-29 17:00Z; dt 0 min

Six-hour averages

Each metric averaged in 6-hour blocks across the stored 72 hours, in UTC. These averages, and the per-site ones below, are what the models were shown.

Source	Metric	Values
asos	humidity	07-28 00Z 85.67, 06Z 89.53, 12Z 71.24, 18Z 55.87, 07-29 00Z 76.33, 06Z 90.25, 12Z 71.09, 18Z 51.84, 07-30 00Z 78.78, 06Z 91.49, 12Z 70.66, 18Z 50.99, 07-31 00Z 75.4
asos	temperature	07-28 00Z 26.3, 06Z 25.4, 12Z 30.09, 18Z 33.68, 07-29 00Z 28.38, 06Z 25.36, 12Z 30.45, 18Z 34.3, 07-30 00Z 28.79, 06Z 26.07, 12Z 30.35, 18Z 34.32, 07-31 00Z 28.94
asos	wind_direction	07-28 00Z 153.8, 06Z 118.1, 12Z 203.8, 18Z 186.3, 07-29 00Z 173, 06Z 85.91, 12Z 191.5, 18Z 193.3, 07-30 00Z 165.2, 06Z 110.7, 12Z 174.6, 18Z 182.7, 07-31 00Z 175.2
asos	wind_speed	07-28 00Z 2.209, 06Z 1.343, 12Z 2.956, 18Z 4.302, 07-29 00Z 3.118, 06Z 0.9042, 12Z 3.001, 18Z 4.181, 07-30 00Z 3.425, 06Z 1.402, 12Z 2.307, 18Z 3.84, 07-31 00Z 3.595
noaa_gfs	gh_500	07-28 06Z 5934, 12Z 5917, 18Z 5931, 07-29 00Z 5923, 06Z 5927, 12Z 5911, 18Z 5929, 07-30 00Z 5924, 06Z 5933, 12Z 5924, 18Z 5940, 07-31 00Z 5940
noaa_gfs	pbl_height	07-28 06Z 357.2, 12Z 278.8, 18Z 1728, 07-29 00Z 1092, 06Z 330.9, 12Z 239, 18Z 2096, 07-30 00Z 1062, 06Z 384.3, 12Z 235.3, 18Z 1652, 07-31 00Z 866.8
noaa_gfs	precipitable_water	07-28 06Z 35.49, 12Z 37.92, 18Z 41.7, 07-29 00Z 42.79, 06Z 36.52, 12Z 39.26, 18Z 43.56, 07-30 00Z 46, 06Z 36.57, 12Z 38.68, 18Z 41.87, 07-31 00Z 44.75
noaa_gfs	surface_pressure	07-28 06Z 1010, 12Z 1010, 18Z 1010, 07-29 00Z 1008, 06Z 1009, 12Z 1010, 18Z 1010, 07-30 00Z 1008, 06Z 1010, 12Z 1012, 18Z 1012, 07-31 00Z 1010
noaa_gfs	t_850	07-28 06Z 21.47, 12Z 20.75, 18Z 18.75, 07-29 00Z 20.13, 06Z 21.27, 12Z 20.59, 18Z 19.02, 07-30 00Z 20.47, 06Z 21.66, 12Z 21.17, 18Z 19.25, 07-31 00Z 21.05
noaa_gfs	u_10m	07-28 06Z 0.8741, 12Z 1.111, 18Z 0.9095, 07-29 00Z -0.3137, 06Z 1.144, 12Z 0.9747, 18Z 1.498, 07-30 00Z 0.2491, 06Z 0.8699, 12Z 1.155, 18Z 0.5909, 07-31 00Z -1.165

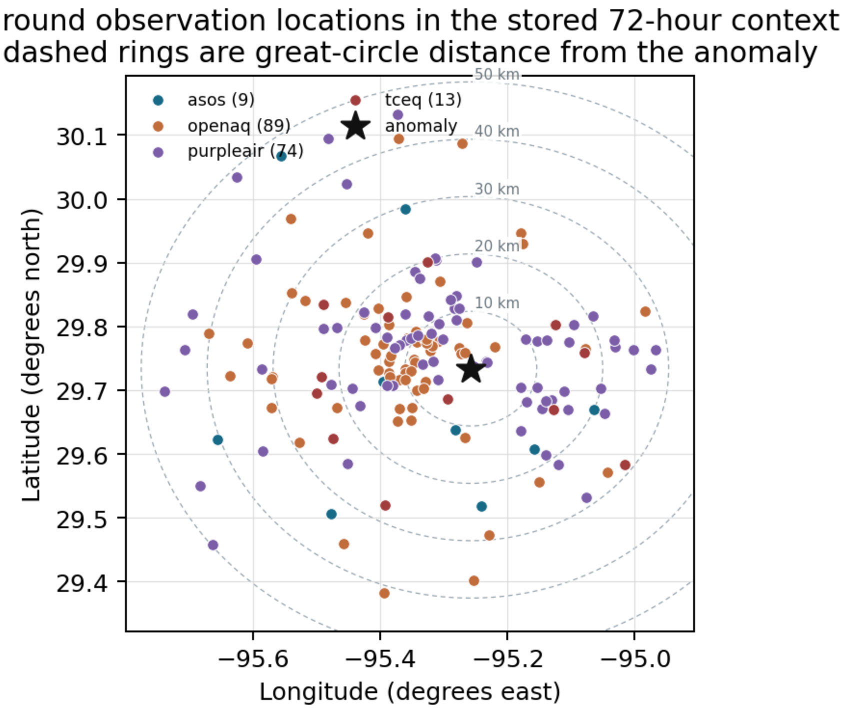
Source	Metric	Values
noaa_gfs	v_10m	07-28 06Z 3.028, 12Z 1.826, 18Z 2.723, 07-29 00Z 4.9, 06Z 2.639, 12Z 1.856, 18Z 2.865, 07-30 00Z 4.345, 06Z 3.252, 12Z 1.391, 18Z 2.493, 07-31 00Z 5.004
openaq	ozone	07-28 00Z 0.01687, 06Z 0.00992, 12Z 0.01452, 18Z 0.02731, 07-29 00Z 0.0169, 06Z 0.008722, 12Z 0.01348, 18Z 0.02853, 07-30 00Z 0.01801, 06Z 0.01094, 12Z 0.01402, 18Z 0.03224, 07-31 00Z 0.01692
openaq	pm10	07-28 00Z 18.67, 06Z 25.28, 12Z 64.11, 18Z 58.17, 07-29 00Z 46.94, 06Z 37.56, 12Z 58.22, 18Z 45.39, 07-30 00Z 26.22, 06Z 22.17, 12Z 51.61, 18Z 37.73, 07-31 00Z 26.78
openaq	pm25	07-28 00Z 5.887, 06Z 5.726, 12Z 5.828, 18Z 5.582, 07-29 00Z 6.283, 06Z 5.394, 12Z 6.89, 18Z 6.695, 07-30 00Z 7.032, 06Z 7.505, 12Z 7.364, 18Z 5.105, 07-31 00Z 5.069
openweather	cloud_cover	07-28 00Z 11.6, 06Z 1.833, 12Z 1.552, 18Z 0.7333, 07-29 00Z 9.161, 06Z 1.367, 12Z 13.83, 18Z 81.55, 07-30 00Z 52.9, 06Z 85.37, 12Z 72.52, 18Z 79.32, 07-31 00Z 86.68
openweather	humidity	07-28 00Z 80.2, 06Z 87.7, 12Z 78.45, 18Z 60.73, 07-29 00Z 73.84, 06Z 88.6, 12Z 78.34, 18Z 58.29, 07-30 00Z 74.53, 06Z 88.77, 12Z 78.69, 18Z 56.52, 07-31 00Z 72.32
openweather	precipitation	07-28 00Z 0, 06Z 0, 12Z 0, 18Z 0, 07-29 00Z 0, 06Z 0, 12Z 0, 18Z 0, 07-30 00Z 0, 06Z 0, 12Z 0, 18Z 0, 07-31 00Z 0
openweather	pressure	07-28 00Z 1012, 06Z 1012, 12Z 1013, 18Z 1011, 07-29 00Z 1010, 06Z 1012, 12Z 1013, 18Z 1011, 07-30 00Z 1012, 06Z 1013, 12Z 1015, 18Z 1013, 07-31 00Z 1014
openweather	temperature	07-28 00Z 27.48, 06Z 26.09, 12Z 29.53, 18Z 35.12, 07-29 00Z 29.97, 06Z 26.18, 12Z 29.85, 18Z 35.49, 07-30 00Z 30.36, 06Z 26.84, 12Z 29.83, 18Z 35.77, 07-31 00Z 30.64
openweather	wind_direction	07-28 00Z 193.6, 06Z 197.4, 12Z 212.3, 18Z 190.2, 07-29 00Z 180.2, 06Z 162.1, 12Z 217.1, 18Z 192.4, 07-30 00Z 175.4, 06Z 186.9, 12Z 233, 18Z 188.2, 07-31 00Z 171.3
openweather	wind_speed	07-28 00Z 3.204, 06Z 2.142, 12Z 2.101, 18Z 2.27, 07-29 00Z 2.553, 06Z 1.714, 12Z 2.068, 18Z 2.284, 07-30 00Z 2.733, 06Z 2.038, 12Z 1.981, 18Z 2.143, 07-31 00Z 2.596
purpleair	pm25	07-28 00Z 7.9, 06Z 7.894, 12Z 6.575, 18Z 6.549, 07-29 00Z 8.031, 06Z 7.668, 12Z 8.763, 18Z 8.366, 07-30 00Z 9.735, 06Z 10.2, 12Z 9.022, 18Z 6.542, 07-31 00Z 7.317
sentinel5p	s5p_aer_ai_granule_available	07-28 18Z 1, 07-29 18Z 1, 07-30 18Z 1
sentinel5p	s5p_ch4_granule_available	07-28 18Z 1, 07-29 18Z 1, 07-30 18Z 1
sentinel5p	s5p_co_column	07-28 18Z 0.02775, 07-29 18Z 0.02655, 07-30 18Z 0.02743
sentinel5p	s5p_co_granule_available	07-28 18Z 1, 07-29 18Z 1, 07-30 18Z 1
sentinel5p	s5p_hcho_column	07-28 18Z 0.0001861, 07-29 18Z 0.0002022, 07-30 18Z 0.0002134
sentinel5p	s5p_hcho_granule_available	07-28 18Z 1, 07-29 18Z 1, 07-30 18Z 1
sentinel5p	s5p_no2_column	07-28 18Z 3.894e-05, 07-29 18Z 3.912e-05, 07-30 18Z 3.936e-05
sentinel5p	s5p_no2_granule_available	07-28 18Z 1, 07-29 18Z 1, 07-30 18Z 1
sentinel5p	s5p_o3_granule_available	07-28 18Z 1, 07-29 18Z 1, 07-30 18Z 1
sentinel5p	s5p_so2_column	07-28 18Z 4.926e-05, 07-29 18Z 6.457e-05, 07-30 18Z 2.458e-05
sentinel5p	s5p_so2_granule_available	07-28 18Z 1, 07-29 18Z 1, 07-30 18Z 1
tceq	co	07-28 06Z 0.1944, 12Z 0.2294, 18Z 0.1778, 07-29 00Z 0.16, 06Z 0.2111, 12Z 0.2722, 18Z 0.1944, 07-30 00Z 0.21, 06Z 0.2, 12Z 0.2313, 18Z 0.1944, 07-31 00Z 0.1778
tceq	no2	07-28 06Z 4.89, 12Z 5.902, 18Z 3.188, 07-29 00Z 2.288, 06Z 5.693, 12Z 5.49, 18Z 3.479, 07-30 00Z 2.292, 06Z 3.799, 12Z 5.327, 18Z 3.942, 07-31 00Z 2.124
tceq	so2	07-28 06Z -0.2111, 12Z 0.08333, 18Z -0.1556, 07-29 00Z -0.09, 06Z -0.2, 12Z -0.1722, 18Z -0.1333, 07-30 00Z -0.1222, 06Z -0.1889, 12Z 0.4278, 18Z -0.1556, 07-31 00Z -0.1444

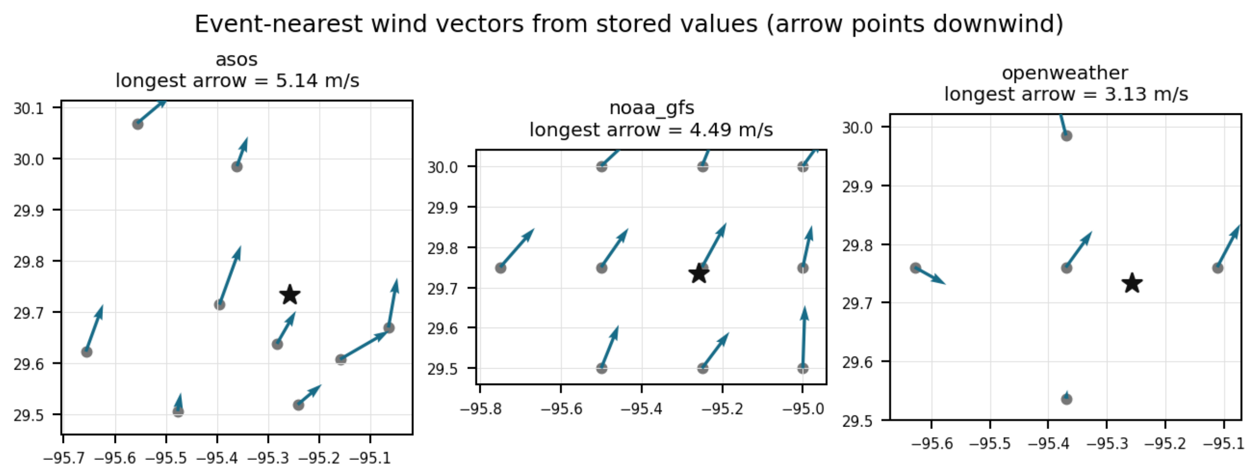
Per-site averages

Each metric averaged at each site, with that site's distance from the event in brackets.

Source	Metric	Values
asos	humidity	HOU (10.965 km) 72.88, MCJ (13.448 km) 71.32, EFD (16.998 km) 73.53, T41 (20.02 km) 74.01, LVJ (23.937 km) 72.78, IAH (29.593 km) 68.4, AXH (33.015 km) 73.07, SGR (40.48 km) 75.97, +1 more
asos	temperature	HOU (10.965 km) 29.85, MCJ (13.448 km) 29.9, EFD (16.998 km) 29.97, T41 (20.02 km) 29.74, LVJ (23.937 km) 29.57, IAH (29.593 km) 30.57, AXH (33.015 km) 28.94, SGR (40.48 km) 29.5, +1 more
asos	wind_direction	HOU (10.965 km) 188.1, MCJ (13.448 km) 182.1, EFD (16.998 km) 173, T41 (20.02 km) 191.7, LVJ (23.937 km) 131.5, IAH (29.593 km) 179.8, AXH (33.015 km) 103.4, SGR (40.48 km) 173, +1 more
asos	wind_speed	HOU (10.965 km) 3.64, MCJ (13.448 km) 3.4, EFD (16.998 km) 3.923, T41 (20.02 km) 3.695, LVJ (23.937 km) 2.111, IAH (29.593 km) 2.654, AXH (33.015 km) 1.209, SGR (40.48 km) 3.426, +1 more
noaa_gfs	gh_500	gfs:29.75,-95.25 (1.952 km) 5928, gfs:29.75,-95.50 (23.473 km) 5928, gfs:29.75,-95.00 (24.937 km) 5927, gfs:29.50,-95.25 (26.0 km) 5927, gfs:30.00,-95.25 (29.617 km) 5929, gfs:29.50,-95.50 (34.993 km) 5927, gfs:29.50,-95.00 (35.994 km) 5926, gfs:30.00,-95.50 (37.722 km) 5929, +2 more
noaa_gfs	pbl_height	gfs:29.75,-95.25 (1.952 km) 966.3, gfs:29.75,-95.50 (23.473 km) 1053, gfs:29.75,-95.00 (24.937 km) 871.5, gfs:29.50,-95.25 (26.0 km) 680.2, gfs:30.00,-95.25 (29.617 km) 1043, gfs:29.50,-95.50 (34.993 km) 778.2, gfs:29.50,-95.00 (35.994 km) 713.8, gfs:30.00,-95.50 (37.722 km) 1165, +2 more
noaa_gfs	precipitable_water	gfs:29.75,-95.25 (1.952 km) 40.44, gfs:29.75,-95.50 (23.473 km) 40.19, gfs:29.75,-95.00 (24.937 km) 40.92, gfs:29.50,-95.25 (26.0 km) 40.43, gfs:30.00,-95.25 (29.617 km) 40.72, gfs:29.50,-95.50 (34.993 km) 40.16, gfs:29.50,-95.00 (35.994 km) 40.51, gfs:30.00,-95.50 (37.722 km) 40.09, +2 more
noaa_gfs	surface_pressure	gfs:29.75,-95.25 (1.952 km) 1011, gfs:29.75,-95.50 (23.473 km) 1009, gfs:29.75,-95.00 (24.937 km) 1011, gfs:29.50,-95.25 (26.0 km) 1011, gfs:30.00,-95.25 (29.617 km) 1009, gfs:29.50,-95.50 (34.993 km) 1010, gfs:29.50,-95.00 (35.994 km) 1012, gfs:30.00,-95.50 (37.722 km) 1007, +2 more
noaa_gfs	t_850	gfs:29.75,-95.25 (1.952 km) 20.42, gfs:29.75,-95.50 (23.473 km) 20.49, gfs:29.75,-95.00 (24.937 km) 20.38, gfs:29.50,-95.25 (26.0 km) 20.43, gfs:30.00,-95.25 (29.617 km) 20.56, gfs:29.50,-95.50 (34.993 km) 20.4, gfs:29.50,-95.00 (35.994 km) 20.53, gfs:30.00,-95.50 (37.722 km) 20.65, +2 more
noaa_gfs	u_10m	gfs:29.75,-95.25 (1.952 km) 0.8703, gfs:29.75,-95.50 (23.473 km) 0.6987, gfs:29.75,-95.00 (24.937 km) 0.4503, gfs:29.50,-95.25 (26.0 km) 0.7012, gfs:30.00,-95.25 (29.617 km) 0.1337, gfs:29.50,-95.50 (34.993 km) 0.202, gfs:29.50,-95.00 (35.994 km) 0.7345, gfs:30.00,-95.50 (37.722 km) 1.551, +2 more
noaa_gfs	v_10m	gfs:29.75,-95.25 (1.952 km) 3.328, gfs:29.75,-95.50 (23.473 km) 3.326, gfs:29.75,-95.00 (24.937 km) 3.057, gfs:29.50,-95.25 (26.0 km) 3.038, gfs:30.00,-95.25 (29.617 km) 2.617, gfs:29.50,-95.50 (34.993 km) 3.421, gfs:29.50,-95.00 (35.994 km) 3.503, gfs:30.00,-95.50 (37.722 km) 2.277, +2 more
openaq	ozone	3378 (0.021 km) 0.01381, 2039 (5.231 km) 0.01818, 325 (6.369 km) 0.005016, 2046 (10.803 km) 0.01553, 320 (12.059 km) 0.01519, 1319333 (14.033 km) 0.02263, 332 (14.34 km) 0.02093, 3214 (14.86 km) 0.02054, +9 more
openaq	pm10	13380082 (0.021 km) 68.4, 15852419 (7.017 km) 25.43, 271 (14.34 km) 30.42
openaq	pm25	15539682 (0.005 km) 8.398, 3379 (0.021 km) 12.61, 8967123 (0.058 km) 9.79, 8967479 (0.058 km) 9.324, 9489522 (2.89 km) 6.762, 8967065 (2.933 km) 7.281, 10252270 (4.015 km) 8.045, 14140751 (6.379 km) 7.068, +61 more
openweather	cloud_cover	grid:center (11.237 km) 39.88, grid:east (14.451 km) 38.6, grid:south (24.535 km) 34.11, grid:north (29.962 km) 48.18, grid:west (35.93 km) 38
openweather	humidity	grid:center (11.237 km) 73.81, grid:east (14.451 km) 78.56, grid:south (24.535 km) 74.86, grid:north (29.962 km) 72.1, grid:west (35.93 km) 74.24
openweather	precipitation	grid:center (11.237 km) 0, grid:east (14.451 km) 0, grid:south (24.535 km) 0, grid:north (29.962 km) 0, grid:west (35.93 km) 0
openweather	pressure	grid:center (11.237 km) 1012, grid:east (14.451 km) 1012, grid:south (24.535 km) 1012, grid:north (29.962 km) 1012, grid:west (35.93 km) 1012
openweather	temperature	grid:center (11.237 km) 30.91, grid:east (14.451 km) 30.19, grid:south (24.535 km) 30.13, grid:north (29.962 km) 30.73, grid:west (35.93 km) 30.35
openweather	wind_direction	grid:center (11.237 km) 204.7, grid:east (14.451 km) 197.1, grid:south (24.535 km) 197.4, grid:north (29.962 km) 170.4, grid:west (35.93 km) 191.5
openweather	wind_speed	grid:center (11.237 km) 2.248, grid:east (14.451 km) 3.615, grid:south (24.535 km) 1.605, grid:north (29.962 km) 2.078, grid:west (35.93 km) 1.601

Source	Metric	Values
purpleair	pm25	99797 (0.6 km) 7.337, 166435 (2.677 km) 10.35, 97279 (2.684 km) 9.325, 6752 (5.276 km) 5.8, 310224 (5.838 km) 9.2, 222601 (6.699 km) 4.718, 133994 (7.377 km) 0.01111, 235425 (8.237 km) 7.336, +66 more
tceq	co	482011035 (0.0 km) 0.1548, 482011039 (14.355 km) 0.1156, 482011052 (15.44 km) 0.3532
tceq	no2	482011035 (0.0 km) 8.636, 482010416 (6.378 km) 4.567, 482011039 (14.355 km) 2.657, 482010026 (14.878 km) 5.762, 482011052 (15.44 km) 11.86, 482011015 (17.437 km) 4.636, 482010024 (19.743 km) 0.3697, 482011066 (22.737 km) 2.608, +4 more
tceq	so2	482011035 (0.0 km) -0.09048, 482011039 (14.355 km) 0.1438, 482010051 (24.236 km) -0.3095





Claims

Mark exactly one box per claim: V (Valid), I (Invalid), or U (Unsure). The labeling guide that comes with this packet has the definitions and marking rules. Each claim appears exactly once, and the number printed next to it is how I'll refer to that claim if I need to ask you about it.

Claim 1 of 36

Sentinel-5P satellite retrievals showed a stable regional CO column density near 0.0265 mol/m2 on July 29, 2026, indicating the anomaly was limited to the surface layer.

Mark exactly one:	[ ] V	[ ] I	[ ] U

Note:

Claim 2 of 36

The data contradict a regional smoke or widespread secondary-pollution explanation because sentinel5p showed a relatively low local carbon monoxide column on 2026-07-29 compared with the adjacent days, ozone near the monitor was only 0.018 ppm, and purpleair and openaq fine-particle levels were modest near the event even though one collocated openaq PM10 sensor was high, indicating a near-source coarse or mixed local plume rather than regional smoke.

Mark exactly one:	[ ] V	[ ] I	[ ] U

Note:

Claim 3 of 36

The PM2.5 levels exceeded the normal range in Houston, Texas on July 29th, indicating a significant increase in particulate matter in the air.

Mark exactly one:	[ ] V	[ ] I	[ ] U

Note:

Claim 4 of 36

OpenAQ sensors co-located near the anomaly reported elevated PM10 concentrations reaching 120.0 ug/m3 at 17:00 UTC on July 29, 2026.

Mark exactly one:	[ ] V	[ ] I	[ ] U

Note:

Claim 5 of 36

The tceq carbon monoxide measurement of 0.5 ppm at 2026-07-29T17:00:00+00:00 exceeded the expected value of 0.13695652173913042 ppm with a z-score of 6.145967130068515, and the anomaly was labeled severe.

Mark exactly one:	[ ] V	[ ] I	[ ] U

Note:

Claim 6 of 36

The anomaly occurred on July 29th at around 17:01:54 UTC, with the nearest sensor located approximately 14.451 km away from the anomaly.

Mark exactly one:	[ ] V	[ ] I	[ ] U

Note:

Claim 7 of 36

TCEQ surface monitors and OpenAQ sensors recorded simultaneous elevations in carbon monoxide reaching 0.5 ppm, PM10 reaching 120 ug/m3, and NO2 reaching 10.5 ppb at 2026-07-29T17:00:00+00:00, supporting a localized industrial or combustion emission source.

Mark exactly one:	[ ] V	[ ] I	[ ] U

Note:

Claim 8 of 36

Regional observations do not support a broad wildfire-smoke or region-wide CO buildup event because Sentinel-5P CO column near Houston was not elevated on 2026-07-29, while the anomalous site simultaneously showed elevated NO2 and high local PM10 consistent with a localized fresh combustion plume.

Mark exactly one:	[ ] V	[ ] I	[ ] U

Note:

Claim 9 of 36

The wind direction and speed on July 29th were conducive to the accumulation of pollutants in the area, leading to the observed air-quality anomaly.

Mark exactly one:	[ ] V	[ ] I	[ ] U

Note:

Claim 10 of 36

Sentinel-5P satellite observations recorded standard baseline total CO column values of approximately 0.0256 mol/m^2 over the area, indicating that the TCEQ carbon monoxide spike was isolated to the surface boundary layer rather than a widespread regional plume.

Mark exactly one:	[ ] V	[ ] I	[ ] U

Note:

Claim 11 of 36

The tceq carbon monoxide measurement at coordinates (29.734, -95.258) was 0.5 ppm at 2026-07-29T17:00:00+00:00.

Mark exactly one:	[ ] V	[ ] I	[ ] U

Note:

Claim 12 of 36

A nearby industrial facility may have released excessive amounts of sulfur dioxide and nitrogen oxides into the atmosphere, causing the air-quality anomaly.

Mark exactly one:	[ ] V	[ ] I	[ ] U

Note:

Claim 13 of 36

The air quality in Houston, Texas is poor due to high levels of NO2.

Mark exactly one:	[ ] V	[ ] I	[ ] U

Note:

Claim 14 of 36

Light winds and stable surface microscale conditions prior to full planetary boundary layer expansion contributed to the localized accumulation of primary combustion emissions, as modeled in GFS data.

Mark exactly one:	[ ] V	[ ] I	[ ] U

Note:

Claim 15 of 36

Localized industrial combustion or flaring events in eastern Houston released primary pollutants, causing concurrent elevated levels of carbon monoxide, nitrogen dioxide, and particulate matter recorded by TCEQ and OpenAQ sensors.

Mark exactly one:	[ ] V	[ ] I	[ ] U

Note:

Claim 16 of 36

A short-lived local combustion source near the Houston monitor, such as industrial process emissions, flaring-related combustion, heavy-duty traffic, or nonroad equipment, is the most plausible physical cause of the severe CO spike because the CO enhancement was large at one TCEQ site while co-pollutants remained only modestly elevated or low at that location.

Mark exactly one:	[ ] V	[ ] I	[ ] U

Note:

Claim 17 of 36

Nitrogen dioxide concentration measured 10.5 ppb at the primary anomaly monitor location on July 29, 2026, at 17:00 UTC.

Mark exactly one:	[ ] V	[ ] I	[ ] U

Note:

Claim 18 of 36

The air quality in Houston, Texas is poor due to high levels of PM2.5.

Mark exactly one:	[ ] V	[ ] I	[ ] U

Note:

Claim 19 of 36

Surface meteorology near the anomaly featured southerly to south-southwesterly winds of about 2 to 2.4 m/s and a deep afternoon planetary boundary layer near 2189 m, which supports transport of a nearby plume to the monitor but does not support strong regional stagnation or shallow-trapping accumulation.

Mark exactly one:	[ ] V	[ ] I	[ ] U

Note:

Claim 20 of 36

A severe carbon monoxide concentration of 0.5 ppm was recorded at coordinates (29.734, -95.258) on July 29, 2026, at 17:00 UTC, significantly exceeding the expected baseline value of 0.137 ppm.

Mark exactly one:	[ ] V	[ ] I	[ ] U

Note:

Claim 21 of 36

NO2 levels were also elevated in Houston, Texas on July 29th, suggesting increased emissions from industrial or vehicular sources.

Mark exactly one:	[ ] V	[ ] I	[ ] U

Note:

Claim 22 of 36

ASOS surface stations and OpenWeather data reported light south-southwesterly winds around 2.0 m/s during the anomaly, providing weak atmospheric dispersion that allowed localized primary pollutants to accumulate near ground level.

Mark exactly one:	[ ] V	[ ] I	[ ] U

Note:

Claim 23 of 36

The temperature was unusually high on July 29th, which could have contributed to the formation of ground-level ozone.

Mark exactly one:	[ ] V	[ ] I	[ ] U

Note:

Claim 24 of 36

The air quality in Houston, Texas is poor due to high levels of SO2.

Mark exactly one:	[ ] V	[ ] I	[ ] U

Note:

Claim 25 of 36

The pollutant showing the anomaly is carbon monoxide measured by tceq.

Mark exactly one:	[ ] V	[ ] I	[ ] U

Note:

Claim 26 of 36

Meteorology from asos, openweather, and gfs supports transport of a localized plume rather than strong stagnation because near the anomaly winds were southerly to south-southwesterly at about 190 to 208 degrees with speeds around 2.1 to 2.4 m/s and the afternoon boundary layer was deep at about 2189 m, allowing advection from upwind source areas while not favoring citywide surface accumulation.

Mark exactly one:	[ ] V	[ ] I	[ ] U

Note:

Claim 27 of 36

The tceq carbon monoxide anomaly is consistent with a localized primary combustion plume because the anomalous monitor measured 0.5 ppm at 17:00 UTC while other nearby tceq carbon monoxide monitors within about 15 km had lower period means of 0.1156 to 0.3532 ppm and the coincident tceq nitrogen dioxide value at the same monitor was elevated at 10.5 ppb, which is a typical co-emitted combustion tracer.

Mark exactly one:	[ ] V	[ ] I	[ ] U

Note:

Claim 28 of 36

The concentration of PM2.5 is elevated to 8.3 ug/m3, which is above the mean value of 8.0665 ug/m3.

Mark exactly one:	[ ] V	[ ] I	[ ] U

Note:

Claim 29 of 36

The pollutant responsible for the anomaly is PM2.5.

Mark exactly one:	[ ] V	[ ] I	[ ] U

Note:

Claim 30 of 36

Heavy urban mobile source emissions from surrounding traffic corridors accumulated locally and were transported toward the monitor by light southerly surface winds reported by ASOS and OpenWeather.

Mark exactly one:	[ ] V	[ ] I	[ ] U

Note:

Claim 31 of 36

A region-wide CO accumulation event or wildfire-smoke-driven anomaly is less plausible because daytime boundary-layer depth was high, no precipitation-related stagnation signal was present, satellite CO column was not elevated over the area, and surface PM2.5 remained only modestly elevated even though one nearby PM10 sensor was high.

Mark exactly one:	[ ] V	[ ] I	[ ] U

Note:

Claim 32 of 36

The TCEQ monitor 482011035 recorded a carbon monoxide concentration of 0.5 ppm at 2026-07-29T17:00:00+00:00, which is a severe positive anomaly relative to the expected value of about 0.137 ppm at that site and time.

Mark exactly one:	[ ] V	[ ] I	[ ] U

Note:

Claim 33 of 36

The air-quality anomaly in Houston, Texas on July 29th was caused by a combination of factors including high temperatures, low wind speeds, and nearby industrial activities.

Mark exactly one:	[ ] V	[ ] I	[ ] U

Note:

Claim 34 of 36

TCEQ monitor 482011035 recorded a carbon monoxide concentration of 0.5 ppm alongside an NO_2 level of 10.5 ppb at 17:00 UTC on July 29, 2026.

Mark exactly one:	[ ] V	[ ] I	[ ] U

Note:

Claim 35 of 36

Coinciding with the carbon monoxide spike on July 29, 2026, at 17:00 UTC, PM10 and PM2.5 concentrations reached 120.0 ug/m3 and 15.7 ug/m3 respectively at a location 0.021 km from the anomaly.

Mark exactly one:	[ ] V	[ ] I	[ ] U

Note:

Claim 36 of 36

Southerly to south-southwesterly near-surface winds of roughly 2 to 3 m/s at the anomaly time could have transported a localized combustion plume from sources south of the monitor toward the site without producing a broad regional pollution episode.

Mark exactly one:	[ ] V	[ ] I	[ ] U

Note:

Optional: most likely cause

Your best guess at the true cause of this event, in your own words. "Insufficient evidence to determine a single cause" is a perfectly fine answer.